11 3 teas and 2 coffees have a total cost of £7.80
5 teas and 4 coffees have a total cost of £14.20

Work out the cost of one tea and the cost of one coffee.

Let t and c be the prices of a tea and a coffee, in pounds.

(1) $3t + 2c = 7.80$

(2) $5t + 4c = 14.20$

1 mark
Form both equations

Double (1): $6t + 4c = 15.60$

Subtract (2): $t = 15.60 - 14.20 = 1.40$

1 mark
Eliminate c by subtracting equal coefficients

Substitute into (1): $3 \times 1.40 + 2c = 7.80$

$$\begin{array}{ccc} 2c & = & 3.60 \\ \div 2 & & \div 2 \\ c & = & 1.80 \end{array}$$

1 mark
Substitute and solve for c

tea £ 1.40

coffee £ 1.80

1 mark
Tea £1.40; coffee £1.80

(Total for Question 11 is 4 marks)